INFO 248 Project 2 Milestone 1 Requirements

Read the doc: INFO 248 Project 2 Final Submission Requirements.pdf so you have an idea of the requirements for the final project submission. You will submit both an ***Rmd file*** and a ***pdf file*** that was knitted from the Rmd file.

Project 2 Required Sections (Milestone 1)

These sections are required:

- **Overview**
    - Give a brief overview of the domain and aspects of that domain that you will study in your analysis. This is to provide context for the reader.
    - State your main research question.
- **Data Description and Processing**
    - Explain where the data set comes from. Include a number from your references section where the link can be found.
    - For each variable,include:
    - The name of the variable.
    - A brief description of what it measures.
    - The data type: numeric or categorical. If categorical, include the names of the levels.
    - Identify the dependent variable.
    - State the size of the data before and after any cleaning, such as removing missing values and/or outliers. Include the count of observations that were removed.
    - Mention if you removed any outliers. State how many were removed.
- **Data Exploration**
    - You include an appropriate (plots,tables) description of the distribution of the outcome variable (categorical or numeric).
    - Provide some text that briefly describes the distribution. The text should be next to the plot and table.
- **Modeling**
  You identify at least two modeling techniques that you plan to use in your analysis.
- **References**
    - List links to your data source accompanied by a number: 1, etc.
    - Use numbers in your report to refer to links in this section, e.g. [1].

Other Requirements

- Your ***dataset*** cannot be one used in this course, and cannot be synthetic (manually generated). It must have more than 50 observations.

- You submit an Rmd file and the pdf file generated from knitting your Rmd file in a single .zip file. See Files to Submit below.
- The metadata section is correct and complete: both names (first and last), title meaningful, date YYYY-MM-DD, using pdf output and number_sections: true.
- The title is descriptive of your analysis. Do not include any reference to “project 2”, etc.
- Do not include any text that refers to this as a class project.
- The number_sections feature is used to appropriately number headings. Use the pound symbols, #, to appropriately organize the level of your headings.
- All library load statements must be in the top code chunk.
- Use the r.version and seed.val variables in any random process in your Rmd file instead of hard-coded numbers. Note: you need to set the random number generator each time you execute a random process.
- No R code or raw output should appear in the knitted pdf file. Use tables, plots and text to summarize output as needed.
- Do not mention any R-specific functions or features in the pdf file. The report should be completely platform agnostic. The reader should not be able to tell what software was used to create the report.
- Examples of how not to mention R-specific terms:
    NO: “...we used the lm function to…”
    YES: “...we fitted a linear regression model…”
    NO: “...we used the na.omit function to remove any NA values…”
    YES: “...we removed any rows that included missing values…”
- All plots must be formatted to fit appropriately on the page, captioned and uniquely numbered.
- All tables must be captioned and uniquely numbered.
- No plots or tables are separated by a page break. Use \newpage to avoid this.

Grading of Milestone 1 (10pts total):

Note: failure to submit a knitted pdf file or an Rmd file that cannot be knitted to pdf will result in a grade of 0.

Formatting	3pts
Analysis	7pts
Total	10pts

Files to Submit

A template Rmd file has been made available for your use. You may use and modify this file as you see fit as long as you fulfill the required sections and other requirements outlined above.

Submit this Rmd file and a pdf file that is the knitted version of the Rmd file. Do not submit any data files. We will only accept an Rmd and a pdf file. Submitting an Rmd file that does not knit will be awarded a maximum of 20% of the project grade.

The files you submit must have this naming convention, replacing the partner1lastname-partner2lastname with your last names:

INFO248-project2ms1-partner1lastname-partner2lastname.pdf
INFO248-project2ms1-partner1lastname-partner2lastname.Rmd

These files must be submitted to the canvas assignment ***in a single archive (zip) file*** named:

INFO248-project2ms1-partner1lastname-partner2lastname.zip

We will not accept any files that do not follow this naming convention. Note: you may use this naming convention for the milestones as well as the final submission.

Plagiarism

We require that your R code and analysis techniques are ***aligned with the examples presented in this class.*** Any variation from these examples ***must be approved by the instructor.*** Submitting code, text, and using other techniques without prior approval will be considered academic dishonesty and may result in severe consequences. Use of AI tools for lookup is OK, but using AI generated output with minimal or no modification is considered plagiarism,an act of academic dishonesty.

Collaboration

You and your partner share the responsibility for the work on the projects in this class. That requires you both to communicate frequently with each other, and to plan the timeline and duties each of you will carry out. If one of you has a particularly challenging week, let your partner know and adjust your work to adapt.

You will individually complete a short assessment of how the collaboration is going before the due date for each project, but if you find that you are having difficulties with your partner, it is important that you contact me directly right away (private piazza post). I will discuss the situation with you in confidence and not take any action without your approval. Please remember that you have a responsibility to your partner to do good work in a timely manner as it will affect both of your grades. ***The instructor reserves the right to modify a project grade based on the level and quality of a student's collaboration.***

Please let the instructor know if you have any questions or concerns about project work at any time.